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1. The North American Plate has a **transform boundary** with the Pacific Plate, dividing California at the San Andreas Fault. Here, the Pacific Plate moves northwest with respect to the North American Plate. Much like in our cookie model, the plates get stuck and, rather than moving smoothly and continuously, they push past each other in short bursts that we feel as earthquakes.
2. Near Alaska, the North American Plate meets the Pacific Plate in a **convergent boundary**, meaning the plates are coming together. Here, the Pacific Plate sinks under the North American Plate in a process called subduction. As the Pacific Plate subducts, some pieces of the crust scrape off onto the North American Plate. However, most of the dense sinking plate gets pulled into the mantle where it melts into the surrounding material. This helps create volcanoes inland from the subduction zone that marks the border between the two plates.